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(54) DATA STORAGE DEVICE AND METHOD OF ACCESS IN CONFIDENTIAL MODE AND NORMAL MODE

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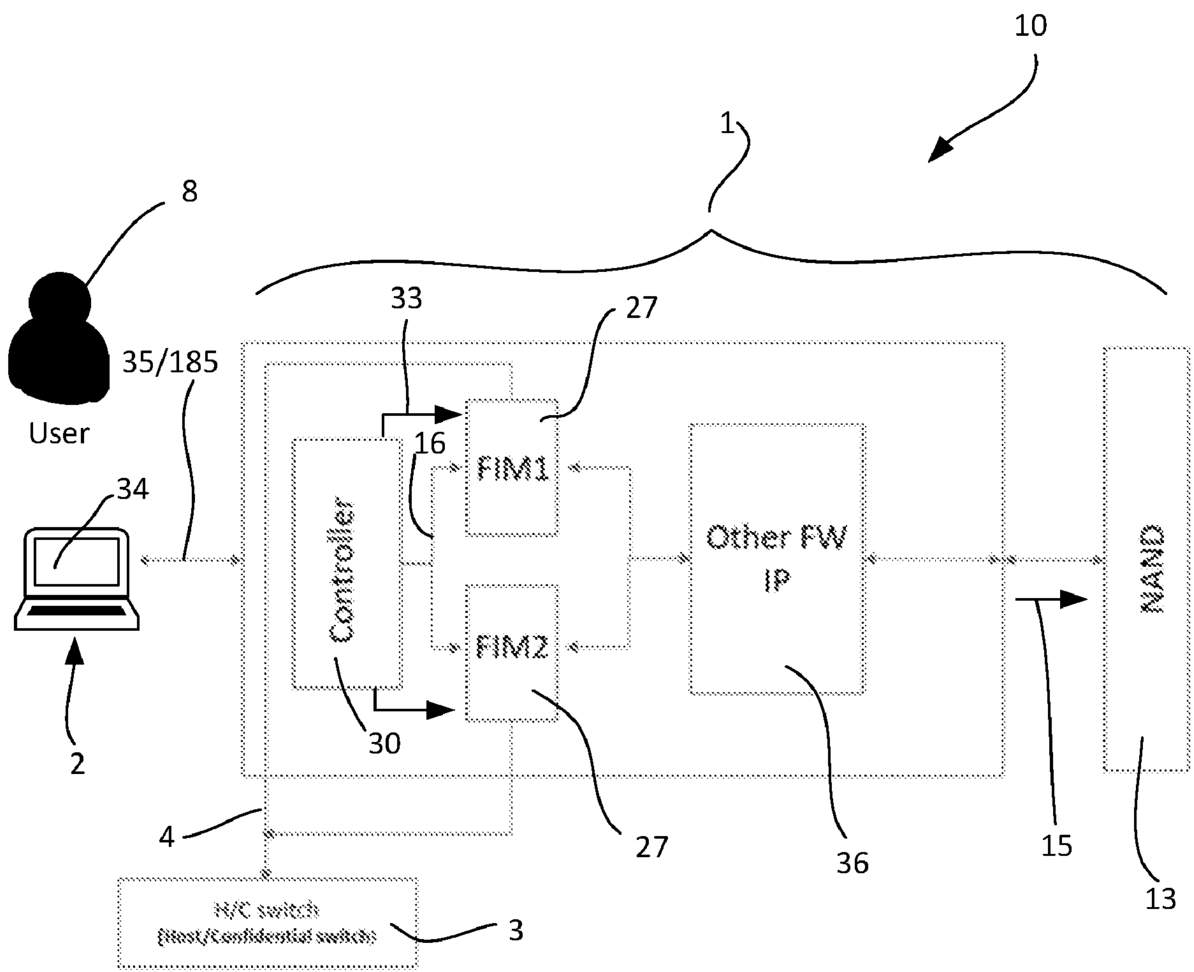
(57) **ABSTRACT**

A data storage device (DSD) (**1**) comprising: a storage medium (**13**); and at least one flash interface module (FIM) (**27**), wherein in response to a signal (**4**) of a switch (**3**) is configured to selectively transition operation of the DSD (**1**) between two modes. A confidential mode, in which security data (**7**) is stored with user data (**9**) as protected data (**11**) in the storage medium (**13**) of the DSD (**1**) during a write operation (**15**), and the security data (**7**) is used to validate subsequent requests to access (**17**), or modify, the corresponding stored user data (**9**). A normal mode, in which access to the protected data (**11**) is prevented.



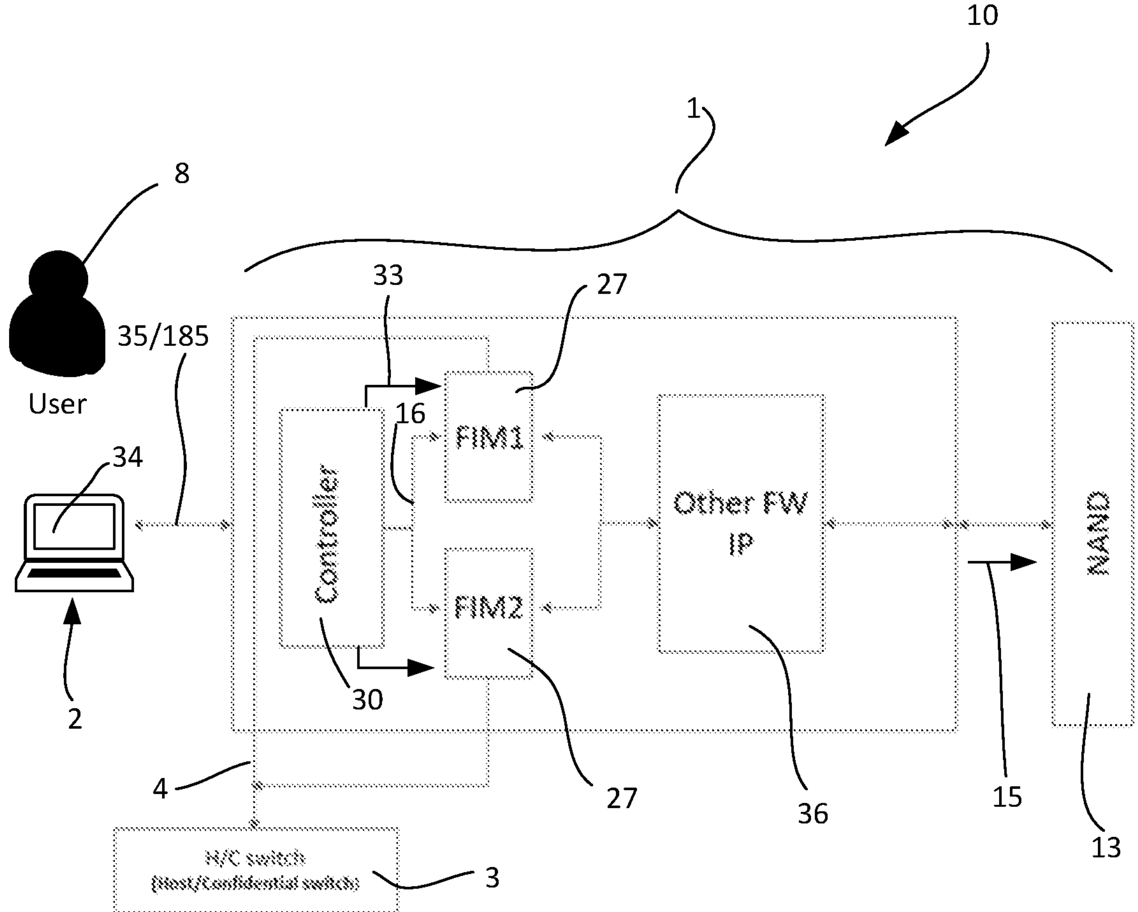

10
1
8
User
35/185
33
27
16
FIM1
Controller
Other FW
IP
NAND
34
FIM2
2
30
4
27
36
15
13
H/C switch
(Host/Confidential switch)
3

Fig. 1

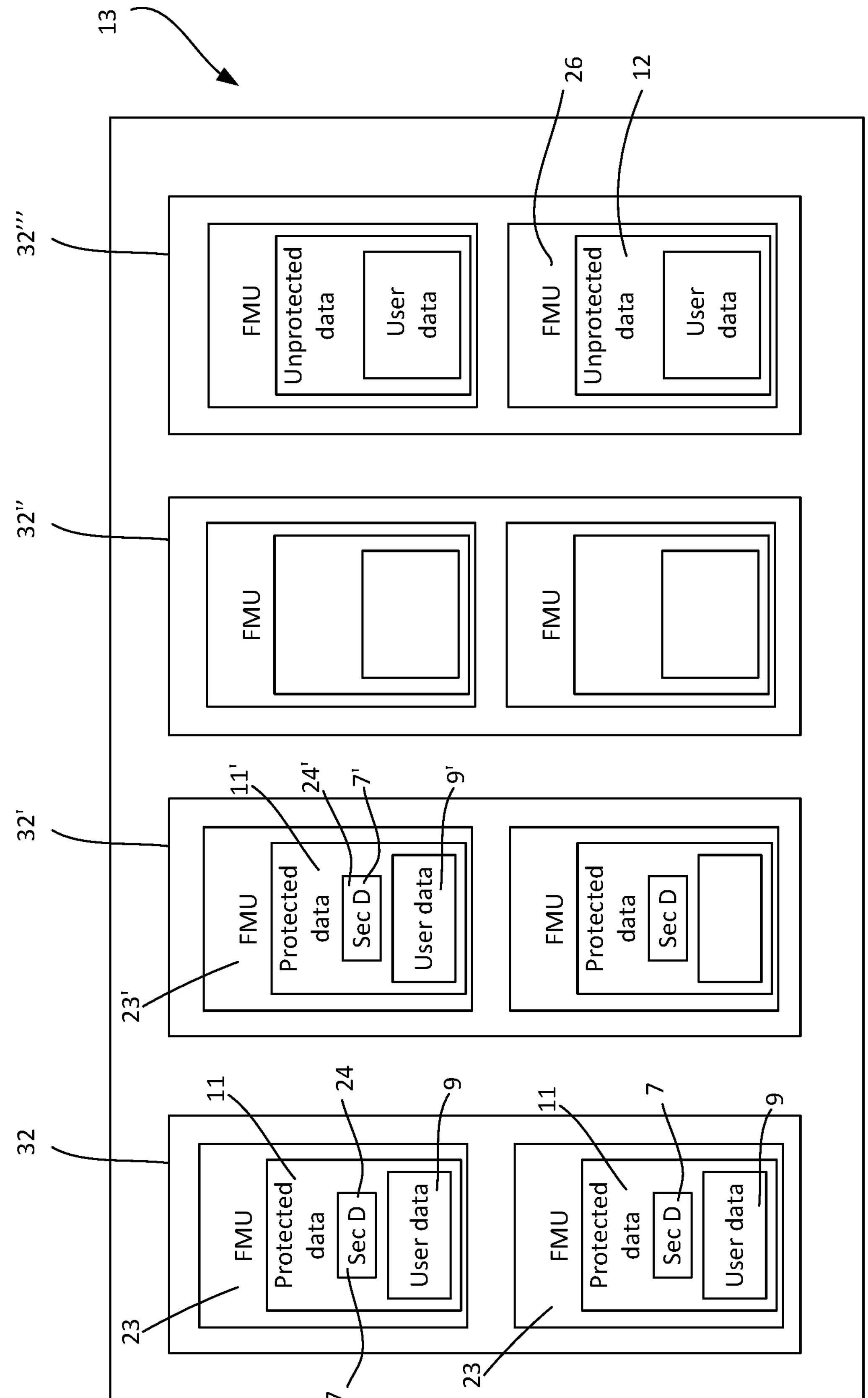

13
32'''
26
12
FMU
Unprotected data
User data
FMU
Unprotected data
User data
32''
FMU
FMU
32'
11'
24'
7'
9'
FMU
Protected data
Sec D
User data
23'
FMU
Protected data
Sec D
32
11
24
9
FMU
Protected data
Sec D
User data
23
7
11
7
9
FMU
Protected data
Sec D
User data
23

Fig. 2

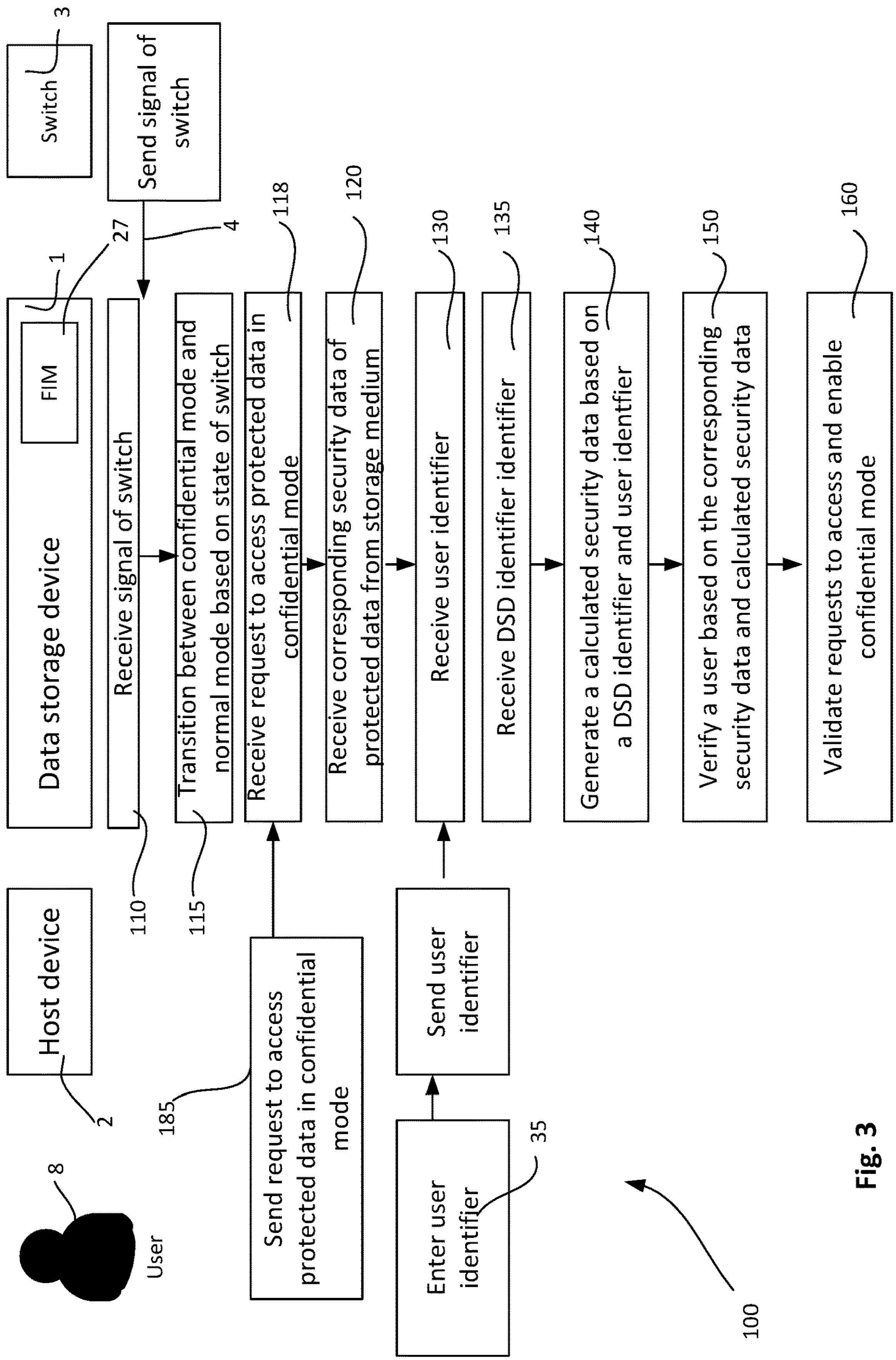

User
8
Host device
2
Data storage device
FIM
1
27
Switch
3
Send signal of switch
4
Receive signal of switch
110
Transition between confidential mode and normal mode based on state of switch
115
Receive request to access protected data in confidential mode
118
Send request to access protected data in confidential mode
185
Receive corresponding security data of protected data from storage medium
120
Receive user identifier
130
Send user identifier
Enter user identifier
35
Receive DSD identifier identifier
135
Generate a calculated security data based on a DSD identifier and user identfier
140
Verify a user based on the corresponding security data and calculated security data
150
Validate requests to access and enable confidential mode
160
100

Fig. 3

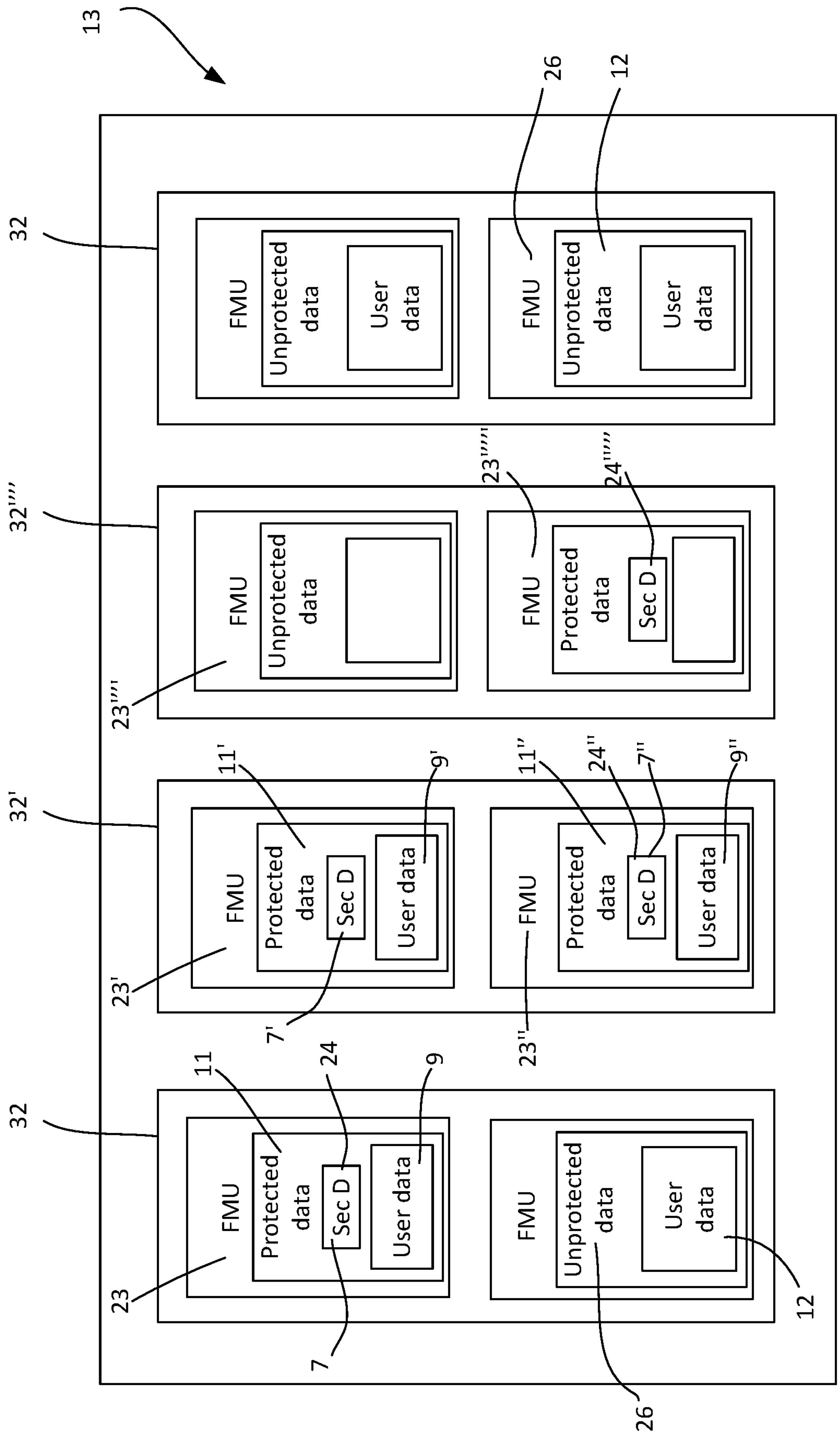

13
32
32'
32'''
32
FMU
Protected data
Sec D
User data
Unprotected data
23
7
11
24
9
26
12
23'
7'
11'
9'
23''
11''
24''
7''
9''
23'''
23''''
24''''

Fig. 4

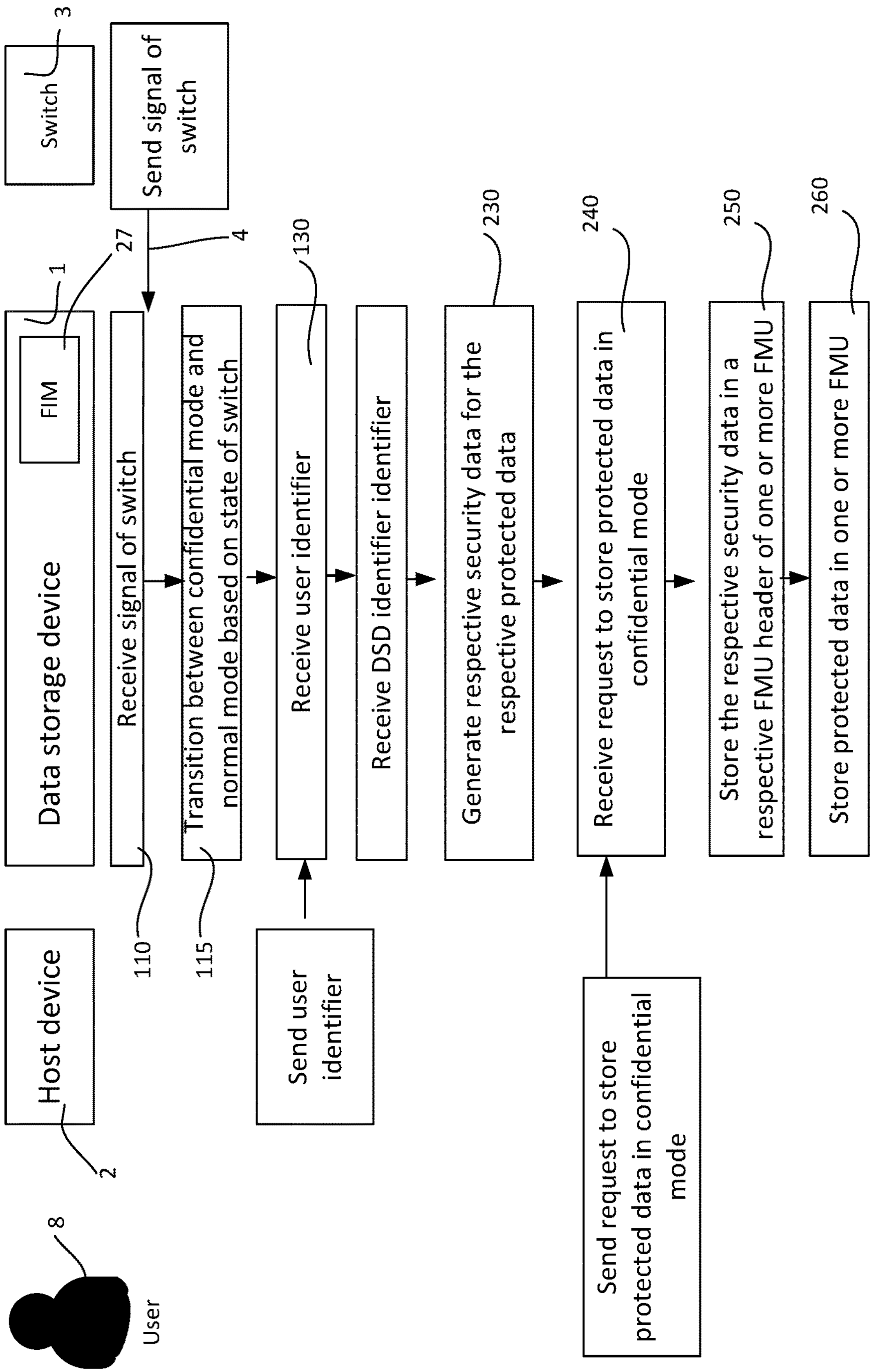

User
8
Host device
2
Data storage device
1
FIM
27
Switch
3
Send signal of switch
4
Receive signal of switch
110
Transition between confidential mode and normal mode based on state of switch
115
Send user identifier
Receive user identifier
130
Receive DSD identifier identifier
Generate respective security data for the respective protected data
230
Send request to store protected data in confidential mode
Receive request to store protected data in confidential mode
240
Store the respective security data in a respective FMU header of one or more FMU
250
Store protected data in one or more FMU
260

Fig. 5

DATA STORAGE DEVICE AND METHOD OF ACCESS IN CONFIDENTIAL MODE AND NORMAL MODE

TECHNICAL FIELD

[0001] The present disclosure relates to a secure data storage device and method of securely accessing and modifying user data.

BACKGROUND

[0002] It is important to protect confidential data from a data storage device. Without protection, one can steal confidential data from a data storage device left unattended by simply copying the confidential data from the data storage device to another storage medium.

[0003] In some examples, the data storage devices can include a storage medium including solid state drive (SSD) using NAND flash memory. SSDs can have firmware that, when operating, by a flash interface module (FIM) manage access to and from a plurality of flash memory units (FMU) of the storage medium.

SUMMARY

[0004] In a first aspect, the present disclosure relates to a data storage device (DSD) comprising: a storage medium); and at least one flash interface module (FIM), wherein in response to a signal of a switch is configured to selectively transition operation of the DSD between a confidential mode and a normal mode. The confidential mode, in which security data is stored with user data as protected data in the storage medium of the DSD during a write operation, and the security data is used to validate subsequent requests to access, or modify, the corresponding stored user data. The normal mode, in which access to the protected data is prevented.

[0005] In some examples of the DSD, the at least one flash interface module (FIM), in response to requests to access, or modify, the protected data from a host device, and in the confidential mode, is configured to: receive the corresponding security data of the protected data from the storage medium; and generate a calculated security data based on a DSD identifier of the DSD and a user identifier provided by the host device; and verify a user based on the corresponding security data being consistent with the calculated security data. In response to verifying the user, the FIM is configured to validate requests to access to enable a data path for the user data between the storage medium and the host device in the confidential mode.

[0006] In some examples, the DSD is configured to store the DSD identifier of the DSD in the storage medium, or another storage medium of the DSD siloed from the storage medium.

[0007] In some examples, the storage medium comprises a plurality of flash memory units (FMU), and wherein each FMU is configurable to store respective protected data, wherein the respective protected data includes respective security data and at least part of the user data.

[0008] In some examples, during a write operation in the confidential mode, the at least one FIM selectively configures one or more of the plurality of FMUs to store respective protected data, wherein the FIM is configured to: generate respective security data for the respective protected data to be stored at the one or more of the plurality of FMUs, wherein the respective security data is based on the DSD identifier of the DSD and the user identifier; store the respective security data in a respective FMU header of the one or more of the plurality of FMUs; and store, at least in part, user data as protected data at the one or more of the plurality of FMUs.

[0009] In some examples of the DSD, the corresponding stored user data as protected data is encrypted, at least in part, with the security data.

[0010] In some examples of the DSD, one or more of the plurality of FMUs is further configurable as an unprotected FMU to store unprotected data.

[0011] In some examples of the DSD, when in the normal mode, the FIM is configured to: disable access to any user data in the one or more plurality of FMU that contains protected data; and enable access to unprotected data in corresponding unprotected FMU.

[0012] In some examples of the DSD, during a write operation in the normal mode the at least one FIM selectively configures one or more of the plurality of unprotected FMUs to store respective unprotected data.

[0013] In some examples of the DSD, the switch comprises: a hardware switch operable by a user; and/or a software switch operable through a user interface of a host device.

[0014] In another aspect, the present disclosure provides a computer-implemented method for selectively enabling access to user data stored in storage medium of a data storage device (DSD), the computer-implemented method performed, at least in part, by at least one flash interface module (FIM) of the DSD, the method comprising: receiving a signal from a switch; wherein in response to the signal, the method includes the DSD transitioning operation between a confidential mode and a normal mode. The confidential mode, in which security data is stored with user data as protected data in the storage medium of the DSD during a write operation, and the security data is used to validate subsequent requests to access or modify the corresponding stored user data. The normal mode, in which access to the protected data is prevented.

[0015] In some examples of the method, wherein in response to requests to access, or modify, the protected data from a host device in the confidential mode, the method comprises: receiving corresponding security data of the protected data from the storage medium; generating a calculated security data based on a DSD identifier of the DSD and a user identifier provided by the host device; verifying a user based on the corresponding security data being consistent with the calculated security data; and in response to verifying the user, validating requests to access to enable a data path for the user data between the storage medium and the host device in the confidential mode.

[0016] In some examples, the method further comprises: receiving, from the host device, the user identifier, wherein the user identifier is a secret associated with the user.

[0017] In some examples, the method further comprises: receiving the DSD identifier of the DSD from the storage medium, or another storage medium of the DSD siloed from the storage medium.

[0018] In some examples, the storage medium comprises a plurality of flash memory units (FMU), and wherein each FMU is configurable to store respective protected data, and wherein the respective protected data includes respective security data and at least part of the user data, wherein in response to requests to access or modify the protected data the method further comprises: identifying one or more FMU that contain the requested protected data; and for each of the identified one or more FMU, the method includes verifying the user based on the respective security data of the FMU being consistent with the calculated security data; and in response to verifying the user validating request to access the protected data in the identified one or more FMU.

[0019] In some examples, wherein during a write operation in the confidential mode, the at least one FIM selectively configures one or more of the plurality of FMUs to store respective protected data, wherein the method further comprises: generating respective security data for the respective protected data to be stored at the one or more of the plurality of FMUs, wherein the respective security data is based on the DSD identifier of the DSD and the user identifier; storing the respective security data in a respective FMU header of the one or more of the plurality of FMUs; and storing, at least in part, protected user data at the one or more of the plurality of FMUs.

[0020] In some examples of the method, wherein storing the protected user data further comprises encrypting the user data, at least in part, with the security data.

[0021] In some examples, the respective security data for each FMU is stored in a corresponding FMU header of the FMU, wherein receiving corresponding security data of the protected data from the storage medium includes receiving the respective security data from the FMU header.

[0022] In some examples, one or more of the plurality of FMUs is further configurable as an unprotected FMU to store unprotected data, wherein in the normal mode, the method further comprises: disabling access to any user data in the one or more plurality of FMU that contains protected data; and enabling access to unprotected data in corresponding unprotected FMU.

[0023] In some examples of the method, wherein during a write operation in the normal mode, the at least one FIM selectively configures one or more of the plurality of the unprotected FMUs to store respective unprotected data, the method further comprising: storing, at least in part, respective unprotected data at the unprotected FMUs.

[0024] In another aspect, the present disclosure provides a data storage device (DSD) comprising: means for storing data; means for receiving a signal from a switch; and means for transitioning operation, in response to the signal, is configured to selectively transition operation of the DSD between: (i) a confidential mode in which security data is stored with user data as protected data in the storage medium of the DSD during a write operation, and the security data is used to validate subsequent requests to access, or modify, the corresponding stored user data; and (ii) a normal mode, in which access to the protected data is prevented.

[0025] In some examples, the data device further comprises: means for receiving requests to access, or modify, the protected data from a host device in the confidential mode; means for receiving corresponding security data of the protected data from the storage medium; means for generating a calculated security data based on a DSD identifier of the DSD and a user identifier; means for verifying a user based on the corresponding security data being consistent with the calculated security data; and in response to the verifying the user, means for validating requests to access, or modify, to enable a data path for the user data between the storage medium and the host device in the confidential mode.

BRIEF DESCRIPTION OF DRAWINGS

[0026] FIG. **1** is a schematic of a system including a data storage device in communication with a host device in accordance with a first example;

[0027] FIG. **2** is a schematic of a storage medium of the data storage device of FIG. **1** including a plurality of flash memory units;

[0028] FIG. **3** is a flow diagram of a method of accessing user data in a data storage device in a confidential mode;

[0029] FIG. **4** is a schematic of a storage medium of the data storage device in an alternative configuration with blocks containing both protected user data and unprotected user data; and

[0030] FIG. **5** is a flow diagram of a method of storing user data in a data storage device in a confidential mode.

DESCRIPTION OF EMBODIMENTS

[0031] In the following detailed description, various aspects of a data storage device in communication with a host device will be presented. These aspects are suited for flash storage devices, such as SSDs (solid-state drive) and SD (Secure Digital) cards. However, those skilled in the art will realize that these aspects may be extended to other types of data storage devices capable of storing data. In yet further examples, the data storage device can be a combination of flash memory and magnetic storage such as a hybrid drive. Accordingly, any reference to a specific apparatus or method is intended only to illustrate the various aspects of the present disclosure, with the understanding that such aspects may have a wide range of applications without departing from the spirit and scope of the present disclosure.

Overview

[0032] FIG. **1** illustrates a system **10** including a data storage device (DSD) **1** that is in communication with a host device **2**.

[0033] The host device **2** and the data storage device **1** may form part of a system **10**, such as a computer system (e.g. server, desktop, laptop, tablet, smartphone, etc.). In this illustrated example, the components of FIG. **1** are physically co-located. However, in other examples the host device may be located remotely from the data storage device.

[0034] The system **10** and method **100** disclosed herein enables the data storage device **1** to selectively operate between a confidential mode and a normal mode. In some examples, a switch **3** is configured to enable a user **8** to selectively transition operation of the DSD **1**. This is illustrated in FIG. **3** where the method includes receiving **110** a signal **4** from the switch **3** and to transition **115** between the confidential mode and the normal mode based on the signal **4**.

[0035] The confidential mode enables user data **9** to be securely stored in a storage medium **13**. In particular security data **7** (as illustrated in FIG. **2**) is stored with user data **9** as protected data **11** in the storage medium **13** during a write operation **15**. The security data **7** is used to validate subsequent requests to access **17**, or modify, the corresponding stored user data **9**.

[0036] In the normal mode, access to the protected data **11** is prevented. In some examples, the normal mode is configured to enable writing of unprotected data, as well as access and modification of such unprotected data by the host device **2**. This can include enabling such writing, access, and modification rights without validating the user of the host device.

[0037] In some examples (as schematically illustrated in FIG. **2**) the storage medium **13** includes flash memory that comprises a plurality of flash memory units (FMU) **23** to store protected data **11** and/or unprotected data **12**. At least one flash interface module (FIM) **27** functions as a multiplexer to enable data paths **16** between the FMU **23** and the host device **2**. Importantly, when a valid request to access **17** is made the FIM **27** is configured to enable a data path **16** for the user data **9** between the FMU **23** and the host device.

[0038] An example of a method of determining a valid request is illustrated in FIG. **3** that shows steps in the method **100** to selectively enable access to the user data **9** in the confidential mode.

[0039] The host device **2** sends **185** one or more request to access the protected data **11** (such as user data **9**) that is received **118** at the data storage device **1**. The method includes receiving **120** corresponding security data **7** associated with the protected data **11** from the storage medium **13**.

[0040] The method also includes generating **140** a calculated security data **31** that is based on a DSD identifier **33** associated with the DSD **1** and a user identifier **35** provided by the host device **2**. The DSD identifier **33**, ideally, is a unique to the DSD **1**. In some examples, the DSD identifier **33** can be a serial number or code, and in other examples this can be a secret code or key. The user identifier **35** may be a secret known to the user(s) **8** (such as a personal identification number or password) who should be authorized to access the user data **9**. Thus the calculated security data **31** can only be generated when there is a combination of a user **8** providing (via host device **2**) the user identifier **35** that is using the particular DSD **1**.

[0041] The method further includes verifying **150** the user **8** based on the corresponding security data **7** being consistent with the calculated security data **31**. In some examples, this may include the corresponding security data **7** being the same as the calculated security data **31** (that is, similar to a shared secret). In other examples, the corresponding security data **7** may be similar to, or share similar characteristics, as the calculated security data **31**. In yet other examples, the corresponding security data **7** may be used to verify a relationship with the calculated security data **31**. For example, the calculated security data **31** may result in a private key and the corresponding security data **7** is a corresponding public key (of a pair of private and public keys). It is to be appreciated other variations of determining consistency, such as other known relationships, can be used to verify the user **8** based on the calculated security data **31** and the stored corresponding security data **7**.

[0042] In some examples, verification of the user based on the security data being consistent with the calculated security data may including subset matching of the security data and calculated security data. This can include multi-user verification techniques suitable for cases where multiple users are sharing the protected data.

[0043] In response to successfully verifying **33** the user **8**, the method includes validating **160** request(s) to access **17** the protected data **11**. This includes enabling a data path **16** for the user data **9** between the storage medium and the host device **2** in the confidential mode.

[0044] In some examples, once the device has been validly switched to the confidential mode the user **8** can continue to operate in the confidential mode for the session without additional verification of the user **8**. That is, it is not necessary to verify the user for every write or read operation. The session may end when the DSD **1** is disconnected (or ejected) or if there is a switch to the normal mode. After this, subsequent use of confidential mode by the user **8** may require further verification of the user **8**.

[0045] The above-mentioned system **10**, data storage device **1**, and method **100** can prevent a nefarious actor who has gained physical control of the data storage device from simply removing the storage medium **13** from the DSD **1** and accessing or copying the data directly from the storage medium **13**. This is prevented in two ways. Firstly, the nefarious actor would not have the user identifier **35**. Secondly, the nefarious actor (and their devices) may not have the DSD identifier **33** to enable the FIM **27** to access the data path.

[0046] In some examples, an advantage of operating at the FIM **27** level to user data at individual FMU **23** is increased granularity, flexibility, and control, and to enable dynamic allocation of memory based on user needs and enabling use for multiple users. That is, enabling users to store as much data (in the confidential mode) into the storage medium as available in the storage medium. This may be in contrast with existing devices, where a user can store data with encryption technologies like SED (self-encrypting drives) or with password protection either for the whole drive or to dedicated logical spaces which needs additional encryption (such as in hardware or software) to enable data privacy.

Data Storage Device

[0047] FIG. **1** illustrates a schematic of a data storage device **1** connected to a host device **2**. The data storage device **1** may be connected to communicate with the host device **2**, such as via a physical data cable **20**. This can include a DSD **1** in the form of a portable device that can be used (at separate times) with more than one host device **2**. In some examples, the DSD **1** may be in wireless communication with the host device **2**, either directly, or via a communications network (not shown). The DSD **1** in particular examples is a portable data storage device utilizing flash memory storage as the storage medium **13**.

Controller 30

[0048] The DSD **1** includes a controller **30** that includes one or more processing devices to perform one or more operations on the DSD **1**. This can include executing instructions from firmware and/or to perform operations on the DSD **1**. This can include functions, such as those of known DSD and SSD controllers such as communication with the host device **2**, wear leveling, error correction, garbage collection, etc.

[0049] The DSD may be configured so that the controller **30** is part of the data path from the FIM **27**/storage medium **13** to the host device **2**.

Switch **3**

[0050] In this example, the DSD **1** includes a physical hardware switch **3** to enable a user **8** to operate and indicate a desired confidential mode or a normal mode. In this example, the signal **4** from the switch is received at the FIM **27**. However it is to be appreciated that the signal **4** indicative of the user selection may be received by the FIM **27** indirectly. For example, the signal **4** from the switch may be sent to the controller **30** that in turn passes a signal to the FIM **27**.

[0051] It is to be appreciated that in some alternatives, the switch **3** may be a software-implemented switch **3** operable via a user interface **34** associated with the host device **2**. In yet another example, the switch **3** may be software-implemented based on other conditions. For example, it may be specified that a particular host device **2** operates in the confidential mode with a specified DSD **1**. Thus DSD **1** may default to operate in the confidential mode (but subject to receiving a valid user identifier **35** to enable access). In other examples, the DSD **1** is configured to operate in a confidential mode on specified time(s) of the day, or on specified date(s), whereby the DSD **1** will attempt to operate in the confidential mode (but also subject the valid user identifier **35**). It is to be appreciated the converse can also be implemented, where alternative conditions result in a software switch **3** to operate the DSD **1** in the normal mode.

[0052] In yet further examples, the software-implemented switch may be triggered (or otherwise controlled) by another hardware switch **3**. For example, the software-implemented switch could be controlled via a wireless interface to another hardware component/switch. In some examples, this may include communication via Bluetooth (including Bluetooth Low Energy protocols), Wi-Fi protocols etc. This can be useful for DSDs that have Bluetooth, Wi-Fi, or other wireless support.

Flash Interface Modules (FIM)

[0053] The DSD **1** also includes one or more flash interface modules (FIM) **27** that interface directly with the FMU **23** in the storage medium **13**. They functionally enable, at least in part, a data path between and the host device **2** as noted above. More specifically, the flash interface modules provide a communication interface between the flash memory units in the storage medium **13** and the controller **30** (which in turn communicates with the host device).

[0054] The disclosed FIM **27**, in addition to functions of known FIM, is operating to provide access control to protected data. The FIM **27** is configured to perform the method **100** described in a separate section below.

[0055] In some examples, the FIM **27** has an input pin in communication with the switch **3** of the drive **3**. This enables the FIM **27** to receive a signal **4** indicative of a selection by a user to use the confidential mode or the normal mode.

[0056] In other examples, the FIM **27** may receive the indication of the selection of the switch **3** via other communication channels. For example, where the switch is based on operation of a user interface **34**, the FIM **27** may receive the signal via the controller **30**, that in turn is in communication with the host device **2**, which in turn is in communication with the user interface **34**.

[0057] When the confidential mode is selected, the FIM **27** is configured to receive DSD identifier **33** of the DSD **1**. In some examples, the DSD identifier **33** is received from the controller **30**. In other examples, the DSD identifier **33** is securely stored in another storage medium that is separate, and siloed, from the storage medium **13** used for user data. This may be advantageous so that if a person removes the storage medium **13**, the DSD identifier **33** cannot be accessed directly from that storage medium **13** which may increase security.

[0058] However in alternative examples, the DSD identifier **33** may be stored in a part of the storage medium **13** that is used for protected data **11**.

[0059] In some examples, as discussed above, the DSD identifier **33** is secret or obfuscated to provide additional security against nefarious actors. However, it is to be appreciate that in alternative examples the DSD identifier **33** is not a strict secret. For example, the DSD identifier **33** is a serial number of the device.

[0060] In further examples, the DSD identifier **33** may be user defined. For example, a user **8** may specify a DSD identifier **33** to be used with the DSD **1**. In yet other examples, each user **9** of a device may assign a respective DSD identifier **33** for the device. In further examples, a user may have a choice of using a default device identifier **33** (such as a serial number) or to specify a DSD identifier **33** to be used. In some examples, the user specified DSD identifier **33** may be encrypted and stored in the DSD.

Encryption Engine **36**

[0061] The DSD **1** may include additional hardware or software encryption engines **36** to add another layer of security. This may include other known data security protocols to obfuscate user data stored in the storage medium. This may be applied when in the confidential mode. In some examples, encryption may also be applied in the normal mode.

[0062] In some examples, the encryption engine **36** may encrypt the user data, at least in part, with the security data **7**.

Storage Medium and Flash Memory Unit (FMU)

[0063] Referring to FIG. **2**. the storage medium **13**, in this example includes flash memory. This may include a plurality of blocks **32**, where each block is the smallest unit that can be erased. Each block contains a plurality of FMU **27**, where the FMU is the smallest data chunk that the can be used to read or write to the flash memory. Because each block is the smallest unit that can be erased, to erase or modify data in one FMU **23**/**26** involves erasing a block **32** and rewriting the block (or to a new block).

[0064] In one example, each FMU **7** may contain up to 4 KB of data (from 8 sectors of 512 bytes for each sector). The storage medium **13** may be packaged in one (or more) units that are mounted to a circuit board via a ball grid array (BGA). In case the storage medium **13** is removed from the device **1** by an unauthorized user, they would not be able to access protected data **11** in the storage medium **13** since they would not have the DSD identifier **33** nor the user identifier **35**.

[0065] Referring to the left-hand side of FIG. **2**, each FMU **23** is configurable to store respective protected data **11**, wherein the respective protected data **11** includes respective security data **7** and protected user data **9**. The security data **7** may be stored in an FMU header **24** of that FMU **23**.

[0066] The FMU **23** may also be configurable as an unprotected FMU **26** to store unprotected data **12** as shown at the block **32'''** on the right-hand side of FIG. **2**. The blocks **32** and FMU **23/26**, when free (as **32''**) can be used for either confidential or normal modes. That is, it is not necessary to predefine or lock in advance which particular memory regions, blocks, or FMU will be used for confidential or normal modes. Instead, the FMU or blocks may be specified on the fly (i.e. as they are written and used).

[0067] FIG. **2** illustrates a "block-based" example where each block **32**, if used in confidential mode, only stores user data for that particular user **8**. Thus the first block **32** has two FMU **23**, where each FMU contains protected data of the same user **8** in the confidential mode. Thus the security data **7** of the two FMU **23** are identical, or are functionally identical, so that the same user identifier **35** can be used to generate calculated security data **31** to enable access to user data **9** of both FMUs **23**.

[0068] If another user (with their own user identifier **35'**) writes to the DSD **1**, then another block **32'** will be used to write the protected data, whereby the respective FMU headers **24'** will be written with security data **7'** based at least in part of that other user's identifier **35''**.

[0069] FIG. **4** illustrates another alternative as a "signature tracking example" where a single block **32** may contain both protected data **11** and unprotected data **26**. The restriction is that each FMU **23**, **26** that is used will contain either protected or unprotected data. In this example, this can increase utilization of memory as it will be possible to write to the block with both unprotected and protected data to fill up the block before opening another block. However, the trade-off is that this can complicate deletion or formatting, when a block contains a mix of unprotected and protected data, or protected data with different users. For example (with reference to left-hand block **32** of FIG. **4**), if the unprotected data **26** needs to be deleted, this cannot be done during normal mode as the entire block **32** will need to be erased which will affect (or require access to) protected data **11** in FMU **23**.

[0070] The second block **32'** of FIG. **4** illustrates a block **32'** with two FMU **23'**, **23''** that holds respective protected data **11'**, **11''** of two users. With the FIM verifying access requests, this enables security of the protected data of different users even if the protected data **11'**, **11''** is saved on the same block **23'**. As there is protected data of two users in the same block, garbage collection process may move the protected data for better block utilization (as will be discussed in further details below).

Host Device and User Identifier **35**

[0071] The host device **2** can be a computer system, computer, laptop, tablet, smartphone, etc. In some examples, the host device **2** can include other electronic devices that is configured to host a data storage device **1**. For example, a smart television, a gaming console, a security camera system, other data recording device, etc. This may be useful for cases where information needs be written and/or accessed securely.

[0072] When the host device **2** is used in the confidential mode, the user identifier **35** needs to be communicated to the DSD **1**. This can include the user **8** entering the user identifier **35** through a user interface **34** associated with the host device **34**. Alternatively, the host device **2** may store the user identifier **35** (such in a digital wallet at the host device **2**). In yet other examples, the user identifier **35** may be stored remotely at a server that can be accessed by the host device **2**. The host device **2** sends the user identifier **35** to the DSD **1**, which can include sending via the controller **30** to the respective FIM **27**.

[0073] In some examples, the user identifier **35** is a secret for a respective user **8**. Thus access to the user data **9** is limited to that user. This can include a user identifier **35** that is a cryptographic key. In some examples, the user identifier **35** may be a combination (or based on a combination) of a public username and a secret password.

[0074] In examples where multiple users wish to collaborate, they may share a common user identifier **35**. That is, the user identifier **35** is shared with multiple users who will use the same user identifier **35** (from their host device) to access the protected data in the DSD **1**.

Modes of Operation

[0075] The two main modes of operation, being the confidential mode and the normal mode, of the data storage device will now be described. This will include initial writing of user data in those modes, followed by access **17** and modifying of the user data.

Confidential Mode

[0076] To enter the confidential mode **5**, the switch **3** is operated to send a signal **4** to the FIM **27**. In some examples, this signal can toggle between the modes. In other examples, the switch can enable a selection of a switch state that corresponds to a confidential or normal mode.

[0077] Actual use of the confidential mode further includes the FIM receiving **135**, **130** valid DSD identifier **33** and a valid user identifier **35**.

Method of Operation Writing in Confidential Mode

[0078] FIG. **5** illustrates an example of writing user data **9** as protected data in the confidential mode. This includes receiving **110** a signal **4** of a switch indicating a user selection of the confidential mode, and subsequent transition **115** to the confidential mode. The method **100** include transitioning **115** to the confidential mode.

[0079] The user **8**, via the host device **2**, sends the user identifier **35** to the DSD **1**, and in turn the FIM. The FIM also receives **135** the DSD identifier **33** of the DSD **1**. This enables the FIM to generate **230** respective security data **7** for the respective protected data **11** to be stored at the one or more plurality of FMUs **27**. The respective security data **7** is based on the DSD identifier **33** and the user identifier **35**.

[0080] The method then includes receiving **240** a request to store protected data from the host device **2**. This request may include the user data **11** to be stored.

[0081] The FIM is configured to store **250** the generated security data **7** in one or more FMU headers **24** of the one or more of the plurality of FMUs **27** that will be storing the user data **9**. The user data **11** is then stored **260** as protected data at the at least one or more of the plurality of FMUs **27**. If the user data **9** has data greater than what can be stored in one FMU, then the user data **9** may be divided up and stored at multiple FMUs **27**. Each FMU that is used to store data in the confidential mode can have a respective FMU header containing respective security data **7**.

[0082] In some examples, user data **9** that is stored as protected data **11** may be further encrypted by encryption engines **36**. In some example, encryption of the user data **9** may be based, at least in part, with the generated security data **7**. This provides an additional layer to prevent a nefarious user from reading the user data **9**.

[0083] As noted above, the blocks of the storage device **13** may be selectively written in different processes.

[0084] In one process, with reference to FIG. **2**, the DSD **1** may be configured so that two types of blocks are open for writing. This can include one block **32'** to enable writing of data in confidential mode and another block **32"** to enable writing in normal mode. When user enables the confidential mode path (for write/read operations), all writes/read will go to the confidential data block **32'** until the user switch back to host normal data mode. Alternatively in the normal mode, the data will be written to data block **32"**. In this way the data written in confidential data mode will not be exposed when device is in host normal data mode. This also helps prevent data deletion when the device is in host normal data mode. In addition deletion (of unprotected data) is easy when device is in host normal data mode without impacting any of the confidential data.

[0085] In another process, with reference to FIG. **4**, the DSD **1** is configured so that only one block is open for all incoming data. Say, for example, block **32""** of FIG. **4** is open to receive data in either the confidential mode or normal mode. When the user enables the host confidential mode, a data path is enabled to write to FMU **23"""** that is then configured with respective security data **7** in the FMU header **24"""**. By having security data in the FMU and having data paths enabled at the FIM/FMU level, this prevents user(s) from accessing and exposing any confidential user data when in the normal mode. This also prevents deletion of user data **9** that is protected data **11** from being deleted in the normal mode. In this alternative process, if the DSD is configured to the normal mode, any data that is written will be stored in the FMU **23""** as unprotected data.

Method of Operation—Access or Modifying Data

[0086] An example method of accessing user data **9** in the confidential mode is illustrated in FIG. **3**, which has been described in the overview above. This includes transitioning to the confidential mode in response to the signal **4** from the switch **3**.

[0087] A user **8** may operate the host device **2** to send **185** requests to access **17** or modify the protected data **11** in the confidential mode. This may include sending the user identifier **35**, as part of the request, or it may be sent from the host device **2** as a separate message.

[0088] On receiving the request to access, the DSD **1** is further configured to identify one or more FMU **23** that contain the requested protected data **11**. This can include identifying from previous data mapping, which particular blocks and FMU **23** contain the requested user data. For each of the identified one or more FMUs **27**, the method includes receiving **120** corresponding security data **7** of the protected data **11**. This can include receiving security data from the FMU header **24**. In this step, the DSD **1** is configured to restrict reading of protected data **11** to only the FMU header **24** (i.e. not allowing reading of user data **9** in that FMU) until after the user is verified **150**.

[0089] The method further includes verifying **150** the user **8** based on the respective security data **7** of the FMU **23** being consistent with the calculated security data **31**. If verified, then the request to access **17** the user data **9** in all the identified FMU **23** will be validated.

[0090] In this example, verifying each FMU from which user data **9** is requested provides additional security. In addition, since verification includes use of a DSD identifier **33** of the DSD **1**, this ensures user data **9** stored in the confidential mode can only be accessed from that specific DSD **1**.

[0091] Once the request to access is validated, the FIM **27** is configured to read user data **9** of FMUs and pass this user data to the host device **2**. If modification of data is required, the user data **9** is read, modified, whereby the user data **8** is then written to a block (similar to the process noted above). In some examples, this includes writing the data **9** to a new block and erasing data in the original block. In yet other examples, this can include temporarily storing the modified user data in memory, then erasing the original block corresponding to the FMU, and then writing the modified user data into the FMU of the erased block.

Normal (Host) Mode

[0092] If the user selects normal mode, the DSD **1** is configured to store data as unprotected data **12** in the DSD **1**. In a write operation in the normal mode, the at least one FIM **27** selectively configures one or more of the plurality of unprotected FMUs **26** to store respective unprotected data. In one example, this can include writing the unprotected data in the FMU **26** as illustrated in FIG. **2**. In alternative examples, when in the host mode a “dummy” user identifier **35**. This dummy may be a generic key that is known, or where the DSD **1** enables any host device to use. This may be advantageous in maintaining a consistent header structure for FMUs in the DSD **1**. In some examples, the normal mode may also use a “dummy” DSD identifier **33**. These dummy identifiers may be stored in the storage medium **13**, or other storage medium (or may be predefined in firmware), or otherwise publicly known.

[0093] When the DSD **1** is in the normal mode, the FIM **27** is configured to disable access to any user data in one or more of the plurality of FMUs **23** that contain protected data **11**. That is, data that was written when the DSD **1** was in a confidential mode. This can be important to prevent cloning of user data in the plurality of FMUs **23** that contains protected data **11**. This includes cloning of drive files containing that protected data as well as cloning of the entire drive (with protected data) to another drive.

[0094] Furthermore, when the DSD **1** is in the normal mode **12**, the DSD may be configured to read, modify (or otherwise access) the unprotected data **12** in any of the corresponding unprotected FMU **26**.

Garbage Collection

[0095] During garbage collection or relocation collection, the FIM **27** may decode the FMU headers to understand the nature of the protected data **11**. Along with the user data **9**, this process copies the security data **7** of the protected data to a destination block **32**. Ideally, the destination block will be storing user data **9** for the same user **8**, and therefore the FMU headers for FMUs will contain the same security data **7**. Hence, even after relocation or garbage collection, the device will retain the confidential information of the file. This was informed by the user during host write operation, and the security data **7** generated at that time stays intact even if the files (of the user data **9**) move around to different memory location in the flash drive.

[0096] This may be useful for situations as shown in the second block **32'** of FIG. **4** where there are two FMU **23'**, **23"** corresponding to two users. The data from these two FMU **23'**. **23"** may be moved to other secured blocks such that each block only holds protected data **11'**, **11"** of one user for better block utilization. In such case, the protected data **11** from second block **32'**, as the source block, is written to the other block where the data includes the user data **9'** with the security data **7'** (whereby the latter is from the FMU header of the source block).

Advantages

[0097] Advantages of the present disclosure include providing a layer of data privacy and security at flash interface module **27** level. This can aid preserving privacy and security of sensitive data even if they have physical control of the storage medium (such as BGA-type NAND flash).

[0098] In some examples, when data is stored in the confidential mode, each flash module unit **23** has a respective FMU header **24** with security data **7**. In addition to increasing to security at the FMU level, this also enables flexibility for users to scale the amount of user data to be stored in the confidential mode. That is, it is not necessary to pre-partition the storage medium with a set region for securing confidential user data.

[0099] In addition, as security data **7** (in some examples) is further based on the data storage device identifier **33**, this can limit access to the protected user data **9** from being accessed by that specific data storage device **1**.

[0100] Furthermore, confidential data can be stored anywhere in the physical memory and there can be preservation of that user data from being transferred from one data storage device to another storage medium. This preservation is enabled by the requirement of requiring verification with the user identifier **35** to prevent unauthorized transfer of user data.

[0101] The disclosed data storage device and method may also enable multiple users (or multiple user profiles) to securely store user data. In some examples, the number of users (or user profiles) are only limited to free memory blocks and respective FMU **23** that are available. The use in confidential mode of one user (with their own security data) will not affect the use of other users from securely storing that other user's confidential data.

[0102] In some examples, the separation of confidential data is provided in at least the block level so that one user cannot access, modify, or erase data of another user since the FIM **27** will bar access to those blocks without valid verification.

Variations

[0103] In some examples, the DSD identifier **33** and/or user identifier **35** may be a unique identifier associated with the DSD **1** and the user **8** respectively. While the DSD identifier **33** and the user identifier **35** can be used by the FIM **27** to identify/verify the specific DSD **1** and user **8**, these identifiers (in some examples) may be obfuscated or kept secret.

[0104] In some examples, the DSD identifier **33** and/or user identifier **35** may include, at least in part, cryptographic keys. That is, having DSD identifier key and user identifier keys. This can include private cryptographic keys to enable authentication. In further examples, these can be used as cryptographic keys to further encrypt and obfuscate user data.

[0105] It will be appreciated by persons skilled in the art that numerous variations and/or modifications may be made to the above-described embodiments, without departing from the broad general scope of the present disclosure. The present embodiments are, therefore, to be considered in all respects as illustrative and not restrictive.

1. A data storage device (DSD) comprising:
a storage medium; and
at least one flash interface module (FIM), wherein in response to a signal of a switch is configured to selectively transition operation of the DSD between:
(i) a confidential mode, in which security data is stored with user data as protected data in the storage medium of the DSD during a write operation, and the security data is used to validate subsequent requests to access, or modify, the corresponding stored user data; and
(ii) a normal mode, in which access to the protected data is prevented.

2. A DSD of claim **1** wherein the at least one flash interface module (FIM), in response to requests to access, or modify, the protected data from a host device, and in the confidential mode, is configured to:
receive the corresponding security data of the protected data from the storage medium; and
generate a calculated security data based on a DSD identifier of the DSD and a user identifier provided by the host device; and
verify a user based on the corresponding security data being consistent with the calculated security data; and
in response to verifying the user, validate requests to access to enable a data path for the user data between the storage medium and the host device in the confidential mode.

3. A DSD according to claim **2**, wherein the DSD is configured to store the DSD identifier of the DSD in the storage medium, or another storage medium of the DSD siloed from the storage medium.

4. A DSD according to claim **2** wherein the storage medium comprises a plurality of flash memory units (FMU), and wherein each FMU is configurable to store respective protected data, wherein the respective protected data includes respective security data and at least part of the user data.

5. A DSD according to claim **4**, wherein during a write operation in the confidential mode, the at least one FIM selectively configures one or more of the plurality of FMUs to store respective protected data, wherein the FIM is configured to:
generate respective security data for the respective protected data to be stored at the one or more of the plurality of FMUs, wherein the respective security data is based on the DSD identifier of the DSD and the user identifier;
store the respective security data in a respective FMU header of the one or more of the plurality of FMUs; and
store, at least in part, user data as protected data at the one or more of the plurality of FMUs.

6. A DSD according to claim **5** wherein the corresponding stored user data as protected data is encrypted, at least in part, with the security data.

7. A DSD according to claim **4**, wherein one or more of the plurality of FMUs is further configurable as an unprotected FMU to store unprotected data.

8. A DSD according to claim **7**, wherein in the normal (host) mode, the FIM is configured to:

disable access to any user data in the one or more plurality of FMU that contains protected data; and

enable access to unprotected data in corresponding unprotected FMU.

9. A DSD according to claim **7**,

wherein during a write operation in the normal (host) mode the at least one FIM selectively configures one or more of the plurality of unprotected FMUs to store respective unprotected data.

10. A DSD according to claim **1** wherein the switch comprises:

a hardware switch operable by a user; and/or

a software switch operable through a user interface of a host device.

11. A computer-implemented method for selectively enabling access to user data stored in storage medium of a data storage device (DSD), the computer-implemented method performed, at least in part, by at least one flash interface module (FIM) of the DSD, the method comprising:

receiving a signal from a switch;

wherein in response to the signal, the method includes the DSD transitioning operation between:

(i) a confidential mode, in which security data is stored with user data as protected data in the storage medium of the DSD during a write operation, and the security data is used to validate subsequent requests to access or modify the corresponding stored user data; and

(ii) a normal (host) mode, in which access to the protected data is prevented.

12. A computer-implemented method according to claim **11**, wherein in response to requests to access, or modify, the protected data from a host device in the confidential mode, the method comprises:

receiving corresponding security data of the protected data from the storage medium;

generating a calculated security data based on a DSD identifier of the DSD and a user identifier provided by the host device;

verifying a user based on the corresponding security data being consistent with the calculated security data; and

in response to verifying the user, validating requests to access to enable a data path for the user data between the storage medium and the host device in the confidential mode.

13. A computer-implemented method according to claim **12** further comprising:

receiving, from the host device, the user identifier, wherein the user identifier is a secret associated with the user.

14. A computer-implemented method according to claim **12** further comprising:

receiving the DSD identifier of the DSD from the storage medium, or another storage medium of the DSD siloed from the storage medium.

15. A computer-implemented method according to claim **12**, wherein the storage medium comprises a plurality of flash memory units (FMU), and wherein each FMU is configurable to store respective protected data, and wherein the respective protected data includes respective security data and at least part of the user data, wherein in response to requests to access or modify the protected data the method comprises:

identifying one or more FMU that contain the requested protected data; and

for each of the identified one or more FMU, the method includes verifying the user based on the respective security data of the FMU being consistent with the calculated security data; and

in response to verifying the user validating request to access the protected data in the identified one or more FMU.

16. A computer-implemented method according to claim **15**, wherein during a write operation in the confidential mode, the at least one FIM selectively configures one or more of the plurality of FMUs to store respective protected data, wherein the method further comprises:

generating respective security data for the respective protected data to be stored at the one or more of the plurality of FMUs, wherein the respective security data is based on the DSD identifier of the DSD and the user identifier;

storing the respective security data in a respective FMU header of the one or more of the plurality of FMUs; and

storing, at least in part, protected user data at the one or more of the plurality of FMUs.

17. A computer-implemented method according to claim **16**, wherein storing the protected user data further comprises encrypting the user data, at least in part, with the security data.

18. A computer-implemented method according to claim **15** wherein the respective security data for each FMU is stored in a corresponding FMU header of the FMU, wherein receiving corresponding security data of the protected data from the storage medium includes receiving the respective security data from the FMU header.

19. A data storage device (DSD) comprising:

means for storing data;

means for receiving a signal from a switch; and

means for transitioning operation, in response to the signal, is configured to selectively transition operation of the DSD between:

(i) a confidential mode in which security data is stored with user data, as protected data, in the storage medium of the DSD during a write operation, and the security data is used to validate subsequent requests to access, or modify, the corresponding stored user data; and

(ii) a normal (host) mode, in which access to the protected data is prevented.

20. A data device according to claim **19** further comprising:

means for receiving requests to access, or modify, the protected data from a host device in the confidential mode;

means for receiving corresponding security data of the protected data from the storage medium;

means for generating a calculated security data based on a DSD identifier of the DSD and a user identifier;

means for verifying a user based on the corresponding security data being consistent with the calculated security data; and in response to the verifying the user, means for validating requests to access, or modify, to enable a data path for the user data between the storage medium and the host device in the confidential mode.

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